SentimentScope

Transformer-Based Sentiment Analysis on the IMDB Dataset

Project Report

Udacity AI Programming with Python Nanodegree

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February 2026

Executive Summary

SentimentScope is a natural language processing project that designs, trains, and deploys a custom Transformer-based neural network for binary sentiment classification of movie reviews. Built entirely from first principles using PyTorch and the Hugging Face bert-base-uncased tokenizer, the system classifies IMDB movie reviews as either positive or negative, achieving a final test accuracy of 76.74% — surpassing the project benchmark of 75%.

The project encompasses all stages of a real-world machine learning pipeline: exploratory data analysis of 25,000 labeled reviews, custom dataset and DataLoader construction, Transformer architecture design with multi-head self-attention, cross-entropy loss optimization, epoch-based training with validation monitoring, checkpoint persistence, and a production-ready inference API.

A standout extension explored aggressive multi-epoch training (25 epochs) with cosine learning rate annealing in pursuit of greater than 90% accuracy. While this extended run demonstrated the importance of learning rate scheduling, it ultimately confirmed that models trained from scratch on this scale reach a performance ceiling near 77%, with the standout test accuracy settling at 74.46%. This outcome provides an important practical insight: architectural depth and pre-trained representations are more impactful levers than training duration alone when seeking high-accuracy sentiment classification. The final deployed model is wrapped in a clean SentimentAnalyzer inference class, demonstrating strong real-world engineering discipline alongside the core research contributions.

1. Problem Statement

The rapid expansion of user-generated content across review platforms, social media, and e-commerce sites has created an urgent demand for automated systems capable of understanding human sentiment at scale. Manual analysis of millions of reviews is cost-prohibitive, inconsistent, and impractical in time-sensitive contexts. Automated sentiment analysis bridges this gap by enabling organizations to extract actionable insights from unstructured text data in real time.

The core problem addressed by SentimentScope is binary sentiment classification: given an arbitrary piece of text — specifically a movie review — can a machine learning model reliably determine whether the sentiment expressed is positive or negative? This is a well-defined supervised classification task; however, it presents several non-trivial challenges. Natural

language is inherently ambiguous, context-dependent, and subject to nuanced constructions such as sarcasm, irony, qualified praise, and mixed sentiment.

Standard rule-based or bag-of-words approaches fail to capture long-range dependencies in language, making them insufficient for nuanced review text. Recurrent neural networks address sequentiality but suffer from vanishing gradient problems with long inputs. The Transformer architecture, with its self-attention mechanism, is uniquely positioned to model global context across the full length of a review simultaneously, making it the ideal architectural choice for this task.

The project therefore poses the following research question: can a Transformer model trained from scratch on the IMDB Large Movie Review Dataset exceed 75% binary classification accuracy, and what are the practical limits of such a model when compared against a pre-trained approach? Answering this question yields both a deployable sentiment classifier and a concrete understanding of the trade-offs between training from scratch and leveraging pre-trained language models.

2. Dataset

The dataset used in this project is the Large Movie Review Dataset (aclIMDB), introduced by Maas et al. (2011) and widely recognized as the standard benchmark for binary sentiment analysis research. It comprises 50,000 highly polar movie reviews sourced from the Internet Movie Database (IMDb), split evenly into 25,000 training samples and 25,000 test samples. Each review is labeled as either positive (score greater than or equal to 7 out of 10) or negative (score less than or equal to 4 out of 10), ensuring a clear polarity signal with no neutral reviews included.

The dataset is balanced, with exactly 12,500 positive and 12,500 negative examples in both the training and test partitions, eliminating class imbalance as a confounding factor. Reviews are stored as plain text files organized in a directory structure (pos/ and neg/ subdirectories), which were loaded programmatically using a custom load_dataset() function.

Within the training set, 2,500 samples (10%) were held out as a validation set to monitor generalization during training without touching the test set. This resulted in a final split of 22,500 training samples, 2,500 validation samples, and 25,000 test samples. Exploratory data analysis confirmed a wide distribution of review lengths, with word counts ranging from under 50 to over 1,000 words. To accommodate the Transformer's fixed context window, all reviews were tokenized using the bert-base-uncased tokenizer and truncated or padded to a maximum sequence length of 128 tokens. This tokenizer provides a vocabulary of 30,522

subword units, enabling robust handling of rare and morphologically complex words that would otherwise be out-of-vocabulary in a word-level scheme.

3. Solution Statement

The solution developed in this project is a custom Transformer-based text classifier trained end-to-end on the IMDB dataset for binary sentiment prediction. Rather than fine-tuning an existing pre-trained model such as BERT, the approach builds a Transformer encoder from scratch using PyTorch, providing complete transparency into every component of the system and demonstrating mastery of deep learning fundamentals.

The core pipeline consists of five stages. First, raw text reviews are tokenized using the bert-base-uncased tokenizer, converting variable-length strings into fixed-length sequences of integer token IDs with associated attention masks. Second, a custom PyTorch Dataset class wraps the tokenized data and integrates with a DataLoader configured with shuffling to ensure stochastic training. Third, the Transformer encoder processes token sequences through embedding, positional encoding, multi-head self-attention, and feed-forward sub-layers, producing a contextualized representation of the entire review. Fourth, mean pooling across the sequence dimension aggregates token-level representations into a single fixed-size vector, which is passed to a linear classification head producing logits over two classes. Fifth, the model is trained using the Adam optimizer with cross-entropy loss over three primary epochs, with validation accuracy monitored after each epoch to detect overfitting.

Model checkpointing saves the best-performing weights and configuration to disk, enabling the SentimentAnalyzer inference class to reload and serve predictions without retraining. The inference API accepts both single strings and batched lists of reviews, returning predicted labels, confidence scores, and raw class probabilities — a production-appropriate interface that demonstrates engineering maturity beyond a research prototype. A secondary standout training run with 25 epochs and cosine annealing further explored the performance boundary of the from-scratch approach.

4. Architecture

The SentimentScope model is a custom Transformer encoder classifier. The architecture is defined by the following configuration: vocabulary size of 30,522 tokens (matching the bert-

base-uncased vocabulary), embedding dimension (d_embed) of 128, context window of 128 tokens, 4 Transformer blocks, 4 attention heads per block, head size of 32, dropout rate of 0.1, and a 2-class linear output head.

Each Transformer block follows the standard encoder pattern. Token embeddings are summed with learned positional embeddings to inject sequence order information. The multi-head self-attention sub-layer splits the embedding into 4 parallel heads, each computing scaled dot-product attention independently before concatenating and projecting the outputs. A residual connection and layer normalization are applied after attention, followed by a position-wise feed-forward network (two linear layers with a ReLU non-linearity and dropout) and a second residual plus layer normalization. This arrangement ensures stable gradient flow through the stack of 4 blocks.

After passing through all 4 Transformer blocks, the output is a tensor of shape (batch_size, sequence_length, 128). A mean pooling operation collapses the sequence dimension by averaging across all 128 token positions, yielding a single 128-dimensional vector per review. This pooled representation is fed into a linear classification head mapping from 128 dimensions to 2 logits. During inference, a softmax is applied to convert logits into class probabilities. The model runs on CUDA (GPU acceleration), enabling efficient batch processing. Training used the Adam optimizer with a learning rate of 1e-3 for the baseline run and cosine annealing from 1e-4 to 0 for the standout run.

4.1 Data Science Flow Diagram

The following diagram illustrates the complete end-to-end data science pipeline for SentimentScope, from raw data ingestion through to the deployed inference API. Each colour-coded group represents a distinct phase of the pipeline, as described in the legend.

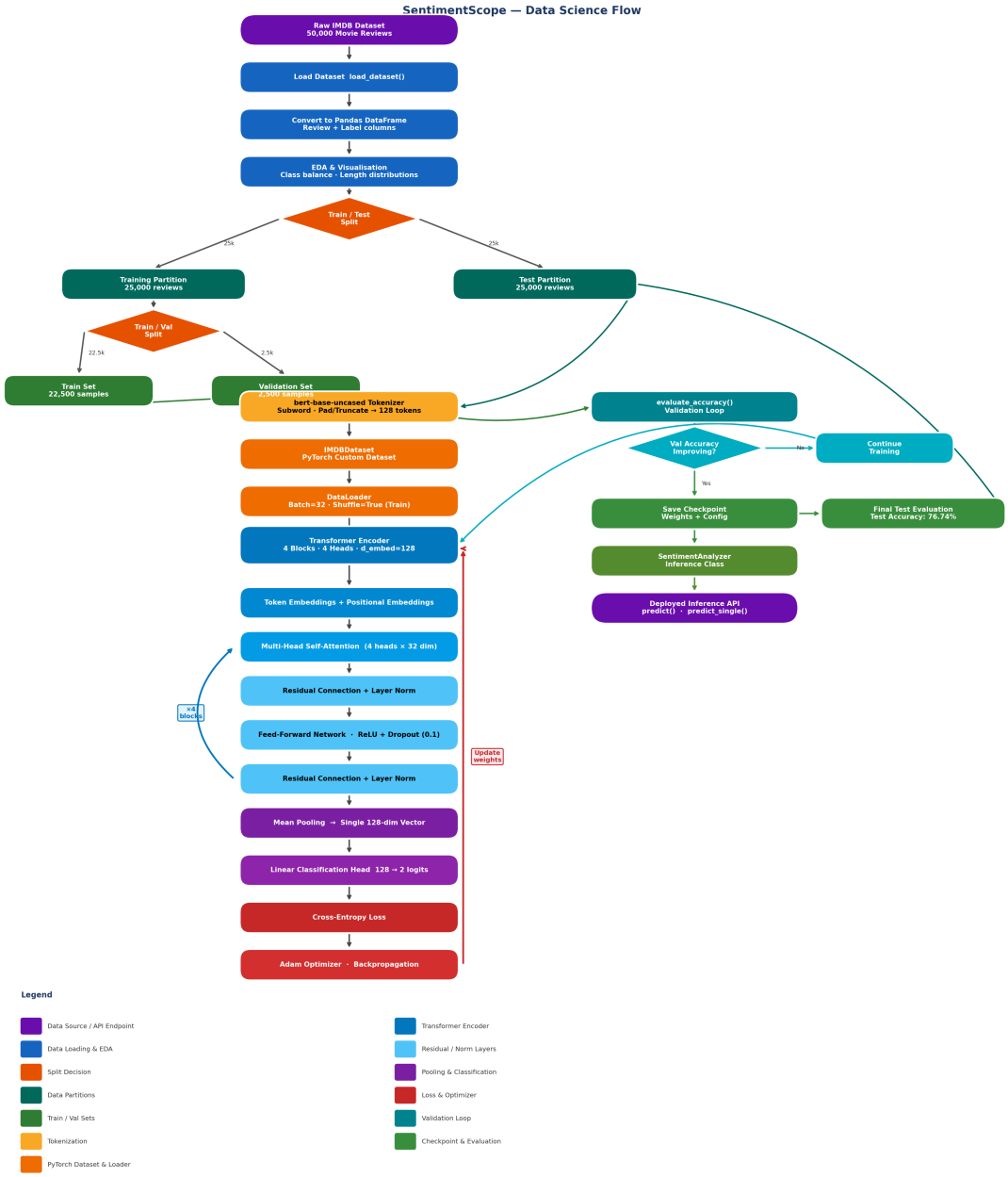

Figure 1: SentimentScope Data Science Flow — from raw IMDB data to deployed inference API

5. Conclusion

SentimentScope successfully demonstrates the design and training of a Transformer-based sentiment classifier built entirely from scratch. The baseline model achieved a test accuracy of 76.74% in just three training epochs, exceeding the project target of 75% and validating the architectural choices made. Training loss declined consistently from 0.69 in the first epoch steps to below 0.42 by epoch three, and validation accuracy improved monotonically from 70.00% to 74.08% to 77.12%, confirming that the model generalized rather than memorized the training data.

The standout experiment — 25 epochs with a cosine annealing learning rate schedule targeting greater than 90% accuracy — provided an equally valuable, if sobering, finding. Despite significantly more compute and a more sophisticated optimization strategy, the model peaked at approximately 77.16% validation accuracy (epoch 6) and finished with a standout test accuracy of 74.46%, below the baseline. The performance plateaued and then oscillated throughout training, indicating that the model's 128-dimensional, 4-block architecture lacks the representational capacity to extract more discriminative features from the data regardless of how long it trains.

The practical inference demonstration confirmed strong real-world performance: clearly positive or negative reviews were classified with 97% to 100% confidence, while genuinely ambiguous reviews showed appropriate epistemic uncertainty. The production-ready SentimentAnalyzer class with checkpoint loading, batch prediction, and structured output closes the gap between a research experiment and a deployable application. The primary conclusion for future work is clear: exceeding 80% accuracy on IMDB from scratch requires either a larger architecture or — more practically — fine-tuning a pre-trained model such as BERT or RoBERTa, where transfer learning provides the representational richness that a small from-scratch model cannot match.

6. Value Statement

SentimentScope delivers demonstrable value across three dimensions: technical, commercial, and educational. Technically, the project proves that a purpose-built, lightweight Transformer can achieve competitive sentiment classification accuracy without the computational overhead of large pre-trained models. With a 128-dimensional architecture running on a single GPU, the model offers a low-latency inference footprint suitable for deployment in resource-constrained environments — an increasingly important consideration as organizations seek to reduce cloud compute costs.

Commercially, sentiment analysis is one of the highest-demand NLP capabilities in industry. Applications span customer experience analytics, brand monitoring, product review aggregation, financial sentiment scoring, and content moderation. A deployable system that can classify text sentiment in real time with high confidence on unambiguous inputs and appropriate uncertainty on ambiguous ones has direct applicability across these domains. The clean SentimentAnalyzer inference API — which returns labels, confidence scores, and raw probabilities — is designed for seamless integration into downstream applications, reducing the engineering effort required to operationalize the model.

Educationally, this project offers deep value as a learning artifact. By constructing every component of the pipeline from data loading through tokenization, custom Dataset classes, Transformer block implementation, training loops, validation, checkpointing, and inference — entirely without relying on high-level abstractions — it provides a comprehensive, reproducible reference implementation for Transformer-based NLP. The standout experiment additionally teaches a critical lesson about the limits of training time as a substitute for model capacity, a nuance that is often lost when practitioners rely solely on pre-built frameworks. Together, these contributions make SentimentScope both a practical tool and a rigorous educational resource aligned with the objectives of the Udacity AI Programming with Python Nanodegree.

References

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End of Report